

The empirical recurrence for OEIS A267953: recovering a conjecture that is not written down

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Abstract

OEIS A267953 records “Empirical recurrence of order 56”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 1688$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 56, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 56 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A267953 is “Number of $n \times 3$ 0..2 arrays with every repeated value in every row and column unequal to the previous repeated value, and new values introduced in row-major sequential order.”. It has offset 1 and begins

4, 96, 1625, 30145, 493087, 8239879, 130870815, 2082478515, ...

The entry states:

Empirical recurrence of order 56 (see link above)

As of the “Last modified” line on the live entry (Jan 22 2016 13:21:15, revision 5) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1688$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1688$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 3376$ exact terms of the model returns order 56 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{56} c_i a(n-i),$$

c_1	35	c_{15}	33445913870844	c_{29}	−11554382972344784249	c_{43}	−100706699165304780
c_2	−143	c_{16}	148180243845746	c_{30}	−14863777269036320891	c_{44}	942816273607316684
c_3	−7496	c_{17}	−496003283868334	c_{31}	23721092883469874076	c_{45}	458979940681134899
c_4	76011	c_{18}	−1526892423404558	c_{32}	34094382910914395591	c_{46}	−12701899446529556
c_5	561151	c_{19}	5339924068786266	c_{33}	−36577499314452940369	c_{47}	−10178363140415160
c_6	−9393361	c_{20}	12156625948621473	c_{34}	−61934983827257414243	c_{48}	−12217632760738611
c_7	−13619720	c_{21}	−42871193998474663	c_{35}	41072519786263254918	c_{49}	16614744806916096
c_8	590391968	c_{22}	−76141903191772771	c_{36}	85696921223372524488	c_{50}	5513757732110336
c_9	−471639190	c_{23}	260497041134418352	c_{37}	−30383040879710454936	c_{51}	−3606605304168448
c_{10}	−22763207864	c_{24}	382045402455563560	c_{38}	−86919104393144410624	c_{52}	1192657925701632
c_{11}	44405467170	c_{25}	−1206961388631076238	c_{39}	9050157257649034016	c_{53}	−165068478087168
c_{12}	590378494527	c_{26}	−1563670944138241224	c_{40}	62453958118927153152	c_{54}	9225589751808
c_{13}	−1555888734407	c_{27}	4274488418554635494	c_{41}	7718297200369446784	c_{55}	−377957122048
c_{14}	−10889130839761	c_{28}	5294046962662408813	c_{42}	−30636755235873672960	c_{56}	−137438953472

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 56, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $56 + 4$ terms removed returns order 56 as well, so $\deg q_{\min} = 56 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $56 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 56 that this sequence satisfies — and that family turns out to have exactly one member.

References

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